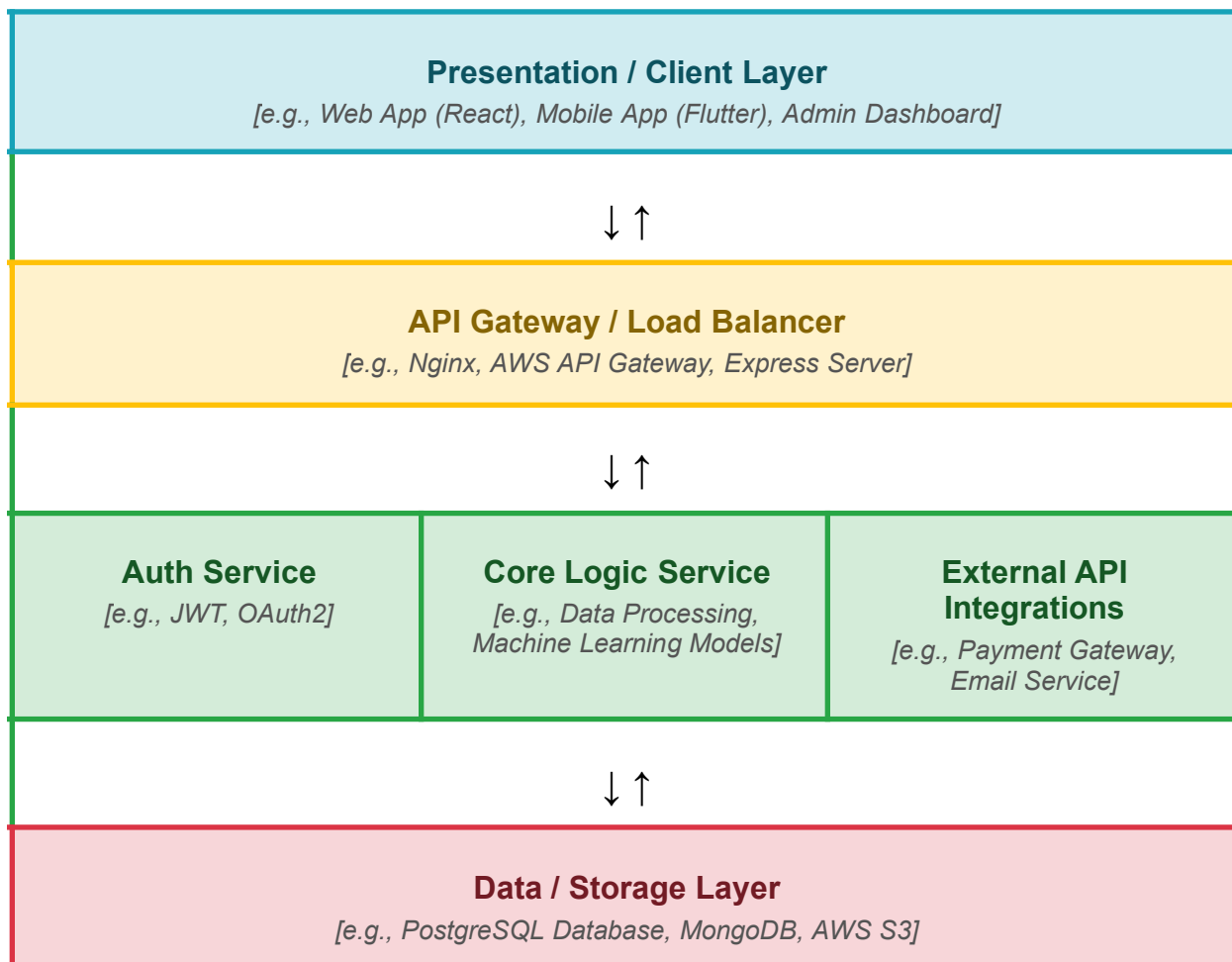


Solution Architecture

Date	30 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Credit Card Approval Prediction System follows a three-layer architecture. The Presentation Layer provides a web interface for users to enter applicant details. The Application Layer uses Flask and a machine learning model to process the input and predict approval status. The Data Layer stores the cleaned dataset, trained model, scaler, and label encoders used for accurate predictions.



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
 Presentation Layer	Provides a web interface for users to enter applicant details (financial, personal, and employment information) and view the credit card approval prediction result.	• HTML, CSS, Bootstrap • Jinja2 Templates • JavaScript
 API Gateway / Application Layer	Receives HTTP requests from the user interface, validates input, and forwards the data to the prediction service. It acts as the entry point and handles routing of requests.	• Flask (Python) • Werkzeug (Routing) • RESTful API (HTTP)
 Microservices (Core Services)	Performs data preprocessing (encoding, scaling), loads the trained machine learning model, and predicts whether the credit card application will be approved or rejected.	• Python • Pandas, NumPy • Scikit-learn • Joblib (Model Loading)
 Database / Data Layer	Stores the cleaned dataset used for training and the essential artifacts required for prediction such as the trained model, scaler, and label encoders.	• CSV (Cleaned Dataset) • Joblib (.pkl files) • Pandas (Data Handling)